



Embedded Systems Development using Zynq

Zynq
Vivado 2014.2 Version

Course Objectives

➤ After completing this course, you will be able to:

- Rapidly architect an embedded system targeting Zynq, and the AXI4 interface standard, using Vivado and IP Integrator
- Extend the system by adding peripherals
 - Add Xilinx provided peripherals from the IP catalog
 - Create and add a custom peripheral using IP Integrator
- Create and debug software applications
 - Create software applications in the Software Development Kit (SDK)
 - Debug an application on-chip using the GNU debugger via SDK

Course Outline

Day 1

The course consists of the following modules:

- **Introduction to Embedded System Design using Zynq and Vivado**
- **Lab 1: Simple Hardware Design**
- **Zynq Architecture**
- **Extending the Embedded System into PL**
- **Lab 2: Adding IPs in Programmable Logic**
- **Adding Your Own Peripheral**
- **Lab 3: Creating and Adding Custom IP**

Course Outline

Day 2

- **Software Development Environment**
- **Lab 4: Writing Basic Software Applications**
- **Software Development and Debugging**
- **Lab 5: Software Debugging Using SDK**

Prerequisites

- Familiarity with the Xilinx tool set and design flow
- Basic C programming
- Basic understanding of processor-based system
- Basic HDL knowledge

Platform Support

➤ Vivado System Edition 2014.2

➤ Xilinx University board

- Zedboard or Zybo

➤ Supported Operating Systems

- Windows XP (SP2) (32/64 Bit)
- Windows 7 Professional (32/64 Bit)
- Windows 8.1 Professional

- Red Hat Enterprise Workstation 5.8-5.10 and 6.4 – 6.5 (32/64 Bit)
- SUSE Linux Enterprise 11.1 – 11.2 (32/64 Bit)
- Cent OS 6.4 6.5 (64-bit)